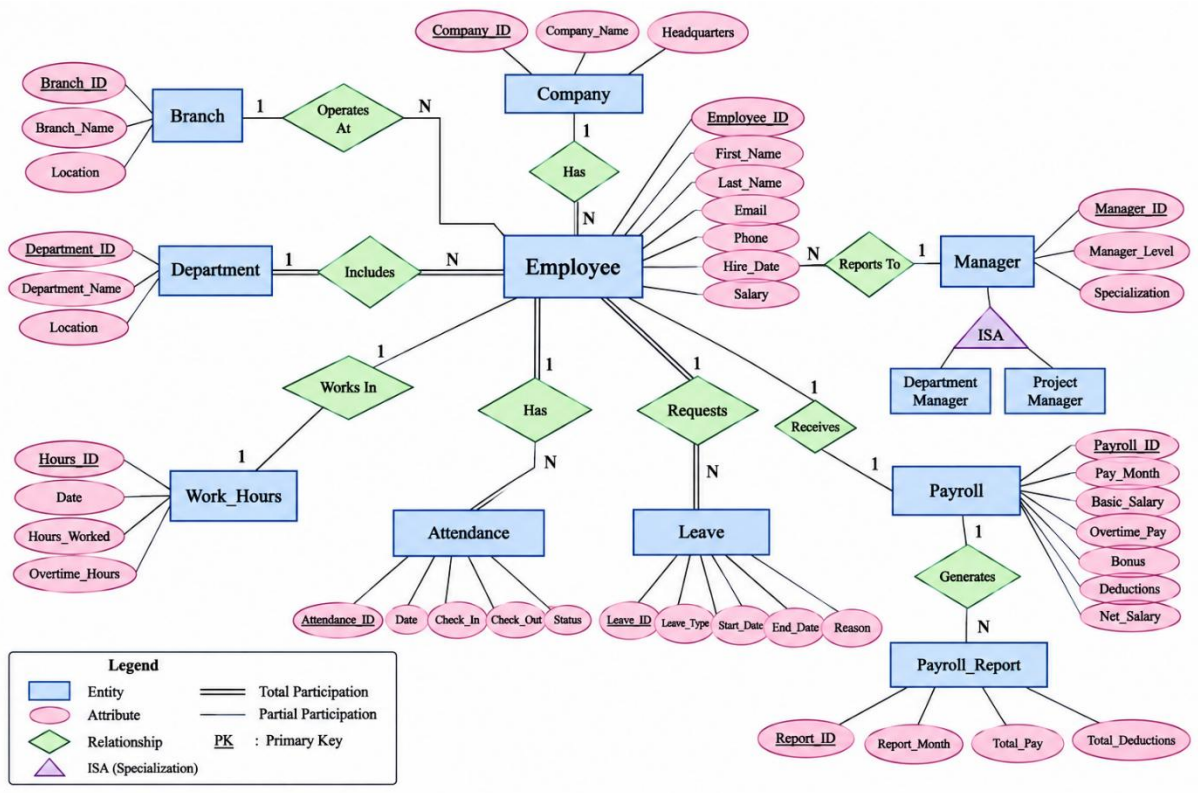
Assignment 1

NAME: RDINESH

REG NO: 192524130

9. Employee Management System Construct an Entity-Relationship (ER) diagram that accurately models the described system. Identify all relevant entities and their attributes, clearly underlining the primary keys. Establish appropriate relationships among the entities and specify the cardinality for each relationship, indicating whether they are one-to-one, one-to-many, or many-to-many. Highlight participation constraints to show whether participation in a relationship is total or partial. Apply generalization or specialization where applicable to simplify and structure the model effectively. Your ER diagram should comprehensively reflect the functional and structural aspects of the system described in the case study and be suitable for conversion into a relational schema and design schema for database and tables A multinational company with several branches across the globe is looking to automate its employee management system to handle employee records, track work hours, manage payroll, and facilitate performance reviews. The system should allow HR managers to add new employees, assign them to specific departments, and track their work hours. Employees should be able to log in to view their personal information, benefits, and payroll details. Managers should have access to performance reviews, team assignments, and leave management. The system should also generate monthly payroll reports, accounting for overtime, bonuses, and deductions. Moreover, the system must handle employee attendance, including time-off requests and leave balances.

DIAGRAM:


Company_ID
Company_Name
Headquarters
Company
Branch_ID
Branch_Name
Location
Branch
1
Operates At
N
1
Has
N
Employee_ID
First_Name
Last_Name
Email
Phone
Hire_Date
Salary
Department_ID
Department_Name
Location
Department
1
Includes
N
Employee
N
Reports To
1
Manager
Manager_ID
Manager_Level
Specialization
ISA
Department Manager
Project Manager
1
Works In
1
Has
1
Requests
1
Receives
Hours_ID
Date
Hours_Worked
Overtime_Hours
1
Work_Hours
N
Attendance
N
Leave
1
Payroll
Payroll_ID
Pay_Month
Basic_Salary
Overtime_Pay
Bonus
Deductions
Net_Salary
1
Generates
N
Payroll_Report
Attendance_ID
Date
Check_In
Check_Out
Status
Leave_ID
Leave_Type
Start_Date
End_Date
Reason
Report_ID
Report_Month
Total_Pay
Total_Deductions
Legend
Entity
Attribute
Relationship
ISA (Specialization)
Total Participation
Partial Participation
PK : Primary Key